

Build PathPilot AI

You are a senior full-stack engineer, AI engineer, and system architect.

Build a production-ready full-stack web application called **PathPilot AI**.

Project Vision

PathPilot AI is an AI-powered Career & Education Advisor platform that helps students discover career opportunities, understand career requirements, identify skill gaps, generate personalized roadmaps, explore government exams, analyze resumes, and receive guidance from an AI mentor.

The platform should be modern, scalable, responsive, and suitable for Smart India Hackathon.

Technology Stack

Frontend:

- Next.js 15
- React
- TypeScript
- Tailwind CSS
- ShadCN UI
- React Hook Form
- Recharts

Backend:

- FastAPI
- Python

Database:

- PostgreSQL

ORM:

- SQLAlchemy

Authentication:

- JWT Authentication

AI/ML:

- Pandas
- NumPy
- Scikit-Learn
- Sentence Transformers

Deployment:

- Docker
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User Roles

Student

Features:

- Register/Login
- Create Profile
- Receive Career Recommendations
- Explore Career Details
- Analyze Skill Gaps
- Generate Career Roadmaps
- Explore Government Exams
- Chat with AI Career Mentor
- Upload Resume

Admin

Features:

- Manage Careers
 - Manage Skills
 - Manage Government Exams
 - Manage Users
 - View Analytics Dashboard
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Required Pages

Landing Page

Sections:

- Hero Section
- Features Overview
- How It Works
- Statistics
- Testimonials
- Call To Action
- Footer

Project Branding:

Title:

PathPilot AI

Tagline:

"Navigate Your Future with Intelligence"

Authentication

Pages:

- Login
- Register
- Forgot Password

Use JWT authentication.

Student Dashboard

Display:

- Profile Summary
 - Career Recommendations
 - Skill Gap Summary
 - Roadmap Progress
 - Recent Activity
 - AI Mentor Widget
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Student Profile

Fields:

- Name
- Age
- Gender
- Education Level
- Stream
- CGPA
- Skills
- Interests
- Certifications
- Career Goals

Allow CRUD operations.

MODULE 1 - CAREER RECOMMENDATION ENGINE

Dataset:

career_recommender.csv

Create a recommendation engine using:

- Skills
- Interests
- Education
- Certifications

Generate Top 5 career recommendations.

Each recommendation must include:

- Career Name
- Match Percentage
- Career Description
- Salary Range
- Future Demand
- Required Skills

Create backend recommendation APIs.

MODULE 2 - CAREER DETAILS EXPLORER

Create detailed career pages.

Each career page should contain:

- Overview
- Eligibility
- Required Skills
- Responsibilities
- Salary Range
- Career Growth
- Industries
- Learning Roadmap

Examples:

- Data Scientist
 - Data Analyst
 - Software Engineer
 - AI Engineer
 - Cybersecurity Analyst
 - Product Manager
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MODULE 3 - SKILL GAP ANALYZER

Input:

Current Skills

Target Career

Output:

- Missing Skills
- Skill Importance
- Suggested Learning Sequence

Example:

Current:

Python

SQL

Target:
Data Scientist

Missing:
Statistics
Machine Learning
Deep Learning

MODULE 4 - ROADMAP GENERATOR

Generate personalized career roadmaps.

Stages:

- Beginner
- Intermediate
- Advanced
- Job Ready

Include:

- Skills
 - Certifications
 - Projects
 - Learning Resources
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MODULE 5 - GOVERNMENT EXAM EXPLORER

Create a searchable exam directory.

Categories:

- UPSC
- SSC
- Banking
- Defence
- Railways
- Teaching
- State PSC
- GATE
- ISRO
- DRDO

Each exam page must contain:

- Overview
- Eligibility
- Age Limit
- Selection Process
- Exam Pattern
- Syllabus
- Salary
- Career Opportunities
- Preparation Resources

Provide search and filtering.

MODULE 6 - AI CAREER MENTOR

Create an AI chatbot.

Capabilities:

- Career Guidance
- Exam Guidance
- Skill Recommendations
- Roadmap Suggestions
- Resume Advice

Example Questions:

- Which career suits me?
- How do I become a Data Scientist?
- What should I learn next?
- Which government exam is suitable after B.Tech?

Store chat history.

MODULE 7 - RESUME ANALYZER

Allow PDF upload.

Analyze:

- ATS Score
- Missing Skills

- Strengths
- Weaknesses
- Career Suggestions

Generate downloadable report.

MODULE 8 - APTITUDE & PERSONALITY ASSESSMENT

Assessment Categories:

- Logical Reasoning
- Quantitative Aptitude
- Verbal Ability
- Personality Assessment

Generate:

- Career Fit Score
- Recommended Career Domains

Store results.

MODULE 9 - ADMIN DASHBOARD

Features:

- Manage Careers
- Manage Skills
- Manage Exams
- Manage Users

Analytics:

- Total Users
- Active Users
- Most Recommended Careers
- Most Viewed Exams
- Skill Trends

Display charts using Recharts.

Database Design

Create PostgreSQL schema for:

users
profiles
skills
interests
careers
career_skills
roadmaps
government_exams
exam_syllabus
resume_reports
chat_history
recommendations
aptitude_results
analytics

Generate SQLAlchemy models and migrations.

API Requirements

Create REST APIs for:

Authentication

Profile Management

Career Recommendation

Career Details

Skill Gap Analysis

Roadmap Generation

Government Exams

Resume Analysis

Chatbot

Admin Dashboard

Include:

- Validation
 - Error Handling
 - Pagination
 - Documentation
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UI Requirements

Design should feel like a modern SaaS product.

Requirements:

- Responsive Design
- Dark Mode
- Dashboard Layout
- Sidebar Navigation
- Global Search
- Professional Education Theme
- Smooth Animations

Use ShadCN components wherever possible.

Development Instructions

DO NOT build everything at once.

Phase 1:

- Project Structure
- Database Schema
- API Architecture
- Frontend Architecture

Phase 2:

- Authentication
- Profile Module

Phase 3:

- Career Recommendation Engine

Phase 4:

- Career Explorer
- Skill Gap Analyzer

Phase 5:

- Roadmap Generator

Phase 6:

- Government Exam Explorer

Phase 7:

- AI Career Mentor

Phase 8:

- Resume Analyzer

Phase 9:

- Admin Dashboard

Generate clean, scalable, production-ready code with proper folder structure, documentation, and best practices.